Blockchain-Based Enrollment Management System for Colegio de Marinduque

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Abstract— This study addresses the inefficiencies of the manual, fragmented enrollment process at Colegio de Marinduque, where students and administrative staff experience delays, redundant verification, and limited record traceability due to the lack of an integrated digital platform. To resolve this, the research proposes a web-based enrollment management system that integrates a microservices architecture with blockchain technology. The system employs SHA-256–based blockchain hashing and record chaining to ensure data immutability, transparency, and tamper detection, while role-based access control (RBAC) secures system operations. Continuous Integration and Deployment (CI/CD) practices further support scalability and maintainability of the platform. Validation results demonstrate that the system performs efficiently, with API response times ranging from 150ms to 320ms across core functions such as registration, enrollment, and record verification. Load testing shows stable system performance up to 250 concurrent users, maintaining low error rates, while accuracy metrics indicate high reliability, with 98.7% precision, 97.9% recall, and an F1-score of 98.3%. Although blockchain integration introduces slight latency, particularly in transaction validation, it significantly enhances data integrity and auditability. The study concludes that combining microservices and blockchain technologies provides a balanced and effective solution for modernizing enrollment systems, improving operational efficiency, ensuring secure record management, and delivering a scalable platform suitable for higher education institutions.

Keywords: *Blockchain, Microservices Architecture, Enrollment Management System, RESTful API, Data Integrity.*

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The enrollment process in educational institutions worldwide remains a complex, multi-step procedure that often relies on traditional, manual methods, leading to inefficiencies and accessibility issues. Globally, enrollment typically involves several stages, such as application, verification, and registration, where participants may drop out at each step due to procedural complexity and administrative burden [1]. In many countries, especially in developing regions, manual enrollment systems still dominate, resulting in long queues, overcrowding, and processing delays. Studies have shown that the absence of digital tools contributes significantly to prolonged waiting times and inefficient service delivery, as institutions struggle to manage increasing student populations and large volumes of records [2], [3]. Additionally, global enrollment challenges are influenced by factors such as population growth, policy implementation, and disparities between urban and rural education systems, further complicating access and management [4].

In the Philippine landscape, enrollment trends reflect both increasing participation and persistent challenges in the education system. According to data from the Philippine Statistics Authority, approximately 32.07 million individuals aged 5–24 years are currently attending school in 2024, while about 10.7 million (25%) remain out-of-school youth (OSCY) [5]. These figures highlight the scale of enrollment demand and the burden placed on educational institutions to efficiently process large numbers of students. The high volume of enrollees, combined with traditional manual enrollment procedures, contributes to long queues, delays, and inefficiencies in record management. Furthermore, disparities in access and administrative capacity across regions underscore the need for improved, digitized enrollment systems to better manage growing student populations and ensure inclusive educational services [6].

Colegio de Marinduque, formerly known as Marinduque Technical and Vocational Institute (MTVI) in Masiga, Gasan, has evolved from a technical-vocational institution established around 2015 into a broader academic institution. Despite this development, its enrollment process remains largely manual and fragmented. Students are required to fill out printed forms, submit physical documents such as report cards, PSA birth certificates, and identification photos, and undergo multiple steps, including aptitude tests and interviews. They must then proceed through various offices such as the registrar's office, the guidance office, and the cashier's office before enrollment is finalized. This multi-step process often results in long queues, repetitive verification, processing delays, and difficulty tracking student records and transactions, particularly during peak enrollment periods.

In response to similar challenges, many higher education institutions have adopted web-based enrollment systems to streamline administrative processes, improve data accuracy, and enhance user experience. These systems digitize key functions such as student registration, course selection, and payment processing, reducing reliance on manual data entry and enabling real-time access to institutional data [7]. However, despite these improvements, traditional centralized systems still present significant limitations. These include vulnerabilities to unauthorized access, susceptibility to single points of failure, and risks of data manipulation, all of which may compromise the integrity and reliability of academic records [8].

To address these limitations, blockchain technology has emerged as a potential solution for secure and transparent data management. As a decentralized and tamper-resistant system, blockchain utilizes cryptographic mechanisms and distributed consensus to ensure that records are immutable, verifiable, and traceable. These features make it particularly suitable for applications that require high levels of data integrity and trust, such as enrollment and academic

record management [7], [9], [10]. By integrating blockchain into enrollment systems, institutions can enhance the security, transparency, and reliability of student data and transactions. However, considerations such as implementation complexity, infrastructure requirements, and user adoption must also be taken into account when applying this technology in an educational setting.

Despite the growing adoption of digital enrollment systems, blockchain-based solutions remain underutilized in higher education institutions, particularly in local and provincial contexts such as Colegio de Marinduque. This gap highlights an opportunity to explore how blockchain technology can be applied to improve existing enrollment processes while addressing concerns related to efficiency, data security, and transparency.

Therefore, this study aims to design and develop a blockchain-based enrollment management system for Colegio de Marinduque. Specifically, it seeks to digitize and simplify enrollment procedures, enhance data integrity and security, improve process efficiency, and evaluate system effectiveness and user acceptability. The findings of this study are expected to contribute to the advancement of secure and efficient enrollment systems in educational institutions.

1.2 Statement of the Problem

The enrollment process at Colegio De Marinduque (formerly Marinduque Technical & Vocational Institute) is inefficient for students and administrative staff due to its manual, fragmented system that requires multiple physical submissions, office visits, and payment steps, leading to delays and redundant verification. The lack of an integrated digital platform limits real-time validation and efficient record tracking, while the absence of blockchain or similar security mechanisms exposes records to tampering, loss, and unauthorized modification, with minimal traceability. This study proposes a web-based enrollment system

with blockchain-backed security to improve data integrity and streamline operations, aiming to achieve a faster, more accurate, and more secure enrollment process during each academic enrollment period.

1.3 Research Questions

Specifically, this study seeks to answer the following questions:

1. How can a blockchain-integrated enrollment management system be developed to digitize and centralize student registration, subject enrollment, payment verification, and academic record management while addressing inefficiencies in manual and fragmented processes?
2. How does blockchain technology improve the security, transparency, and immutability of enrollment records and transactions compared to traditional systems?
3. How effective and acceptable is the proposed system to end-users in terms of efficiency, accessibility, accuracy, usability, security, and overall performance?

1.4 Objectives of the Study

- **General Objective:** The project aims to develop a Blockchain-Integrated Enrollment Management System for Colegio de Marinduque to streamline student registration, subject enrollment, payment verification, and academic record management.

- **Specific Objectives:**

    1. To design and develop a blockchain-integrated web-based enrollment management system that centralizes student registration, subject enrollment, payment processing, and academic record management while automating manual processes.
    2. To implement blockchain technology to ensure secure, transparent, and immutable handling of enrollment records, payment transactions, and student data.

3. To evaluate the system's effectiveness and acceptability using ISO/IEC 25010 standards and user acceptance testing in terms of functionality, usability, efficiency, security, and overall performance.

1.5 Significance of the Study

This study proposes a blockchain-integrated web-based enrollment management system designed to improve the efficiency, security, and reliability of enrollment processes at Colegio De Marinduque. The system aims to address the limitations of the current manual, fragmented procedures by providing a centralized, automated platform for enrollment operations. The following stakeholders are expected to benefit from the study:

The **Enrollment Advisers** will benefit from automated subject advising, including real-time validation of prerequisites, curriculum alignment, and detection of schedule conflicts. This reduces manual workload, minimizes human error, and allows more efficient handling of student enrollment concerns.

The **Students** will benefit from a more convenient and accessible enrollment process through online subject selection, real-time schedule availability, and immediate access to billing and academic records. The system reduces the need for physical document submission and long queues while ensuring secure and transparent transactions.

The **Cashier** will benefit from improved payment processing through automated fee calculation, accurate transaction recording, and faster payment verification. Blockchain integration ensures that all financial records are secure, immutable, and traceable, reducing discrepancies and enhancing financial accountability.

The **Registrar** will benefit from a centralized, secure student record management system that improves the storage, retrieval, and updating of academic and enrollment records. The use of blockchain technology ensures data integrity, prevents unauthorized modifications, and provides a reliable audit trail for all transactions.

Overall, the study contributes to modernizing the enrollment system at Colegio De Marinduque by improving operational efficiency, strengthening data security, and enhancing service delivery for all stakeholders.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Review of Related Literature

2.1.1 Blockchain Technology in Educational Administration

Blockchain technology has emerged as a transformative innovation in educational administration, providing a secure, transparent, and decentralized framework for managing academic records, credentials, and institutional processes. Nadry et al. [11] highlight its role in enhancing transparency and accountability in both educational management systems and public administration through tamper-proof credential verification, efficient credit transfer, and long-term preservation of academic records. These capabilities foster trust among stakeholders—including students, institutions, employers, and government agencies—while enabling auditable governance and improved service delivery.

Complementing this perspective, blockchain is widely recognized as a foundational technology for Education Evaluation Management (EEM), where it functions as a distributed ledger that ensures data integrity, reduces reliance on centralized authorities, and addresses issues such as credential fraud, lack of auditability, and administrative inefficiencies [12]. Mechanisms such as verifiable credentials and self-sovereign identity further empower learners to control and securely share their academic data. In addition, smart contracts automate evaluation processes—including grading policies and credential issuance—thereby improving efficiency and reducing administrative overhead [12].

From an architectural standpoint, blockchain-based educational systems commonly adopt hybrid models in which sensitive data are stored off-chain while cryptographic proofs are anchored on-chain. This approach ensures scalability, integrity, and privacy, aligning with regulatory frameworks such as the GDPR, which emphasize data minimization and controlled disclosure [13]. Certificate issuance and verification remain the most prominent applications,

addressing concerns about fraudulent credentials and enabling reliable cross-institutional validation. User-friendly verification tools such as QR codes and web-based platforms further enhance accessibility [13].

Despite these advantages, challenges persist, including high implementation costs, scalability limitations, privacy risks, technical complexity, and resistance to organizational change, as well as the need for clear governance and regulatory frameworks [11], [12]. Overall, blockchain holds significant potential to modernize educational administration; however, its successful implementation requires a balanced integration of technology, policy, and institutional readiness.

2.1.2 Blockchain for Academic Credential Management and Verification

Blockchain-based academic credential systems are increasingly conceptualized as layered, privacy-aware infrastructures that balance verifiability, scalability, and regulatory compliance. These systems anchor credential data on blockchain through tamper-evident hashes or metadata, while storing full academic records off-chain to address storage limitations and comply with privacy regulations such as GDPR [14]. This hybrid architecture enables secure, decentralized verification without exposing sensitive personal information, often supported by tools such as QR codes and web platforms.

The literature emphasizes key principles, including data minimization, selective disclosure, and the use of private or consortium blockchains to align with governance requirements. Moreover, blockchain directly addresses the issue of fraudulent academic credentials by ensuring immutability, traceability, and independent verifiability through cryptographic proofs [9]. Effective implementations also consider interoperability across institutions, long-term data persistence, and resilience when issuing entities cease operations.

Recent studies further highlight sustainability concerns, demonstrating that energy-efficient consensus mechanisms such as Proof-of-Authority (PoA) are more suitable for educational environments than resource-intensive approaches like Proof-of-Work (PoW) [10]. Despite these benefits, challenges remain in achieving standardization, ensuring regulatory compliance across jurisdictions, and scaling infrastructure. Overall, blockchain provides a robust framework for enhancing trust and efficiency in credential management, but requires integration with governance frameworks and privacy-preserving architectures for effective implementation [14], [15], [16].

2.1.3 Integration of Blockchain and Microservices in Enrollment Management Systems

Microservices architecture has evolved as a distributed system design approach in which applications are decomposed into small, independent services that operate and communicate through lightweight APIs. This architectural style improves scalability, flexibility, and maintainability compared to traditional monolithic systems [17]. Although not originally designed for enrollment systems, these principles are highly applicable to academic platforms requiring modular, scalable service delivery, particularly during high-demand periods such as student enrollment.

In enterprise environments, microservices emphasize service independence, decentralized data management, and API-driven communication, enabling systems to achieve improved performance and fault isolation [18]. These characteristics are essential in enrollment management systems, where multiple processes—such as student registration, course selection, and record verification—must operate concurrently without causing system-wide delays. The separation of these functions into independent services supports efficient processing and reduces bottlenecks.

Furthermore, microservices architectures support continuous delivery and horizontal scalability, allowing systems to handle increasing workloads by scaling individual components rather than the entire application [21]. This is particularly relevant in academic settings where user demand fluctuates, especially during enrollment periods. By distributing workloads across services, the system can maintain stable performance even under concurrent access conditions.

On the other hand, blockchain technology has been widely recognized for its ability to ensure data integrity, transparency, and security in distributed systems. Blockchain operates on a decentralized ledger, where transactions are validated by consensus mechanisms and secured with cryptographic techniques, preventing unauthorized data modification [19]. This makes it highly suitable for managing sensitive academic records and enrollment transactions.

In addition, blockchain has been identified as an effective solution for secure record management and verification across various domains, including education [20]. Its ability to provide immutable and traceable records addresses common issues in traditional systems, such as data tampering, lack of transparency, and reliance on centralized control. Although existing studies do not specifically focus on enrollment systems, the application of blockchain to academic record management provides a strong foundation for integrating it into enrollment processes.

Overall, current literature does not present a fully integrated system that combines both microservices architecture and blockchain technology for enrollment management. However, existing studies consistently demonstrate that microservices enhance system performance and scalability, while blockchain strengthens security and data integrity. These complementary strengths justify integrating both technologies into the proposed system, providing a more efficient, secure, and scalable solution than traditional enrollment management systems.

2.1.4 Blockchain-Based Enrollment Systems and Credential Management in Higher Education

Blockchain technology is increasingly recognized as a viable solution for modernizing enrollment systems and enabling cross-institutional academic data management. Credential verification remains the most mature application, supporting secure, transparent, and tamper-evident handling of academic information [22]. Building on this capability, systems such as BCHEEN demonstrate how blockchain can facilitate scalable, interoperable enrollment processes by addressing inefficiencies in fragmented, centralized systems [23].

These systems employ hybrid architectures that store sensitive data off-chain while anchoring verification proofs on-chain, ensuring privacy, integrity, and scalability in alignment with regulatory frameworks [14]. Permissioned blockchain networks further enhance governance and access control, enabling secure collaboration among institutions without reliance on centralized intermediaries. Features such as selective disclosure, tokenized credit systems, and user-friendly verification tools expand system functionality and usability.

Empirical evaluations indicate that such systems can achieve acceptable performance while reducing storage overhead and administrative burden [23]. However, challenges remain, including limited storage capacity, interoperability issues, regulatory compliance, and the need to ensure long-term data availability. Adoption is also uneven, with many systems still in pilot stages. Overall, blockchain-enabled enrollment systems present a promising direction for transforming higher education administration, though their success depends on addressing technical, regulatory, and governance challenges [22].

2.1.5 Data Privacy Compliance in the Philippine Academic System

While blockchain enhances data integrity and trust, its application in education must align with privacy regulations, security requirements, and governance frameworks. Blockchain is best understood as a socio-technical system, in which risks extend beyond cryptographic

mechanisms to encompass vulnerabilities in consensus protocols, smart contracts, and institutional governance structures [24]. To address these concerns, most educational systems adopt hybrid architectures in which sensitive academic data are stored off-chain while cryptographic proofs are recorded on-chain. This approach supports verifiability while minimizing exposure of personally identifiable information and aligns with global privacy principles such as data minimization and selective disclosure [13], [25].

In the Philippine context, data privacy compliance is governed by the Data Privacy Act of 2012, which mandates the protection of personal information across all institutions, including higher education [31]. The National Privacy Commission provides specific guidelines for educational institutions, emphasizing lawful data processing, consent, accountability, and the implementation of both technical and organizational safeguards [32]. These regulatory requirements impose important design constraints on blockchain-based systems, particularly because blockchain's immutability may conflict with principles that allow data to be modified or erased. As a result, privacy-preserving strategies—such as off-chain storage, encryption, and controlled data access—are essential to ensure compliance within the Philippine legal framework.

Beyond technical design, effective data privacy compliance requires strong governance and institutional readiness. Research indicates that many higher education institutions encounter challenges in implementing comprehensive data governance frameworks, including inconsistent policy enforcement, limited technical infrastructure, and varying levels of data privacy awareness among stakeholders [25]. In blockchain-based systems, governance mechanisms such as permissioned networks, role-based access control, and audit trails play a crucial role in ensuring accountability and regulatory compliance across institutions. Smart contracts can further enhance operational efficiency by automating credential processing and

verification; however, they must be carefully audited to prevent vulnerabilities and ensure alignment with legal standards.

Despite these advancements, several challenges remain, including harmonizing international and local regulatory frameworks, ensuring interoperability across diverse systems, and maintaining long-term data accessibility, particularly when issuing institutions cease operations. Additionally, balancing transparency with privacy remains a critical issue, especially in cross-institutional environments where secure data sharing is required. Overall, blockchain-based education systems must adopt a privacy-by-design approach that integrates hybrid data architectures, robust governance models, and strict compliance with regulations such as RA 10173. This ensures not only secure and efficient academic data management but also ethical and legally compliant implementation within the Philippine educational landscape [13], [24], [25], [31], [32].

2.1.6 Challenges and Adoption of Blockchain in Higher Education

The adoption of blockchain technology in higher education is influenced by technical, institutional, and socio-regulatory challenges that affect its scalability and sustainability. Despite its advantages in transparency, security, and decentralization, adoption remains limited due to infrastructural constraints, lack of institutional readiness, and policy gaps [27], [29].

Technical challenges include scalability issues, interoperability with legacy systems, and energy consumption associated with certain consensus mechanisms [23], [30]. Furthermore, integrating blockchain into existing institutional frameworks requires organizational change and coordination across multiple stakeholders, which can be difficult to achieve [29].

Privacy and regulatory concerns further complicate adoption. Compliance with data protection laws such as GDPR necessitates the use of hybrid architectures, encryption, and

selective disclosure mechanisms to protect sensitive information [28]. Additionally, user trust and acceptance are critical, as stakeholders must be confident in the system's security and legal validity [28].

The literature emphasizes the need for standardized frameworks, governance models, and institutional policies to support interoperability and consistent implementation [23], [29]. Overall, while blockchain holds transformative potential, its successful adoption depends on addressing technical limitations, enhancing institutional readiness, and establishing clear governance and regulatory frameworks [16], [27], [28], [29], [30].

2.2 Review of Related Systems

2.2.1 Blockchain-Based E-Voting System for Universities (Consortium Blockchain Model)

Blockchain-based e-voting systems in universities are typically implemented using a consortium (permissioned) blockchain architecture, where only authorized participants—such as administrators, validators, and eligible students—can interact with the system. In the implementation proposed by Marouan et al. [33], smart contracts encode election rules directly into the blockchain, enabling automated enforcement of voting procedures without manual intervention. The architecture follows a multi-layered structure composed of a user interface (web or mobile), secure API communication, and a blockchain backend that records transactions immutably. Cryptographic techniques such as SHA-256 hashing and ECDSA digital signatures are employed to ensure vote integrity and voter authentication [33]. Furthermore, Delgado-von Eitzen et al. emphasize the importance of privacy-aware design, recommending off-chain storage with cryptographic proofs to prevent exposure of sensitive personal data while maintaining verifiability [13].

This architectural approach offers several advantages. It ensures a tamper-evident and auditable voting process, as all transactions are permanently recorded and independently verifiable [33]. The consortium model enhances governance by restricting participation to trusted institutional entities, thereby reducing security risks. Additionally, cryptographic mechanisms strengthen data integrity and voter authenticity, while privacy-preserving techniques support compliance with data protection regulations [13]. These features make blockchain-based e-voting particularly suitable for academic environments where transparency and controlled participation are critical.

However, the architecture also presents several limitations. Compliance with data privacy regulations necessitates hybrid on-chain and off-chain storage models, increasing system complexity and reliance on external components [13]. Blockchain latency may affect real-time vote processing due to consensus mechanisms [33]. Integration with existing university systems, such as student information systems and authentication platforms, also poses technical challenges. Moreover, consortium blockchains introduce governance risks, as system security depends on the coordination and trustworthiness of validator nodes [33]. Overall, while the model provides strong security and transparency, careful architectural design is required to address performance, privacy, and integration concerns.

2.2.2 ICP Blockchain-Based Student Voting System (Web3 DApp Architecture)

The Internet Computer Protocol (ICP)-based student voting system represents a decentralized Web3 architecture that supports secure, transparent, and programmable student interactions. In the framework proposed by Rao et al. [34], the system uses ICP's decentralized infrastructure to manage voting processes, student records, and token-based transactions, all secured by cryptographic authentication and consensus mechanisms. This reflects a broader

trend in educational systems toward adopting Web3 principles such as decentralization, programmability, and tokenization.

The architecture incorporates secure identity management, enabling users to authenticate with cryptographic credentials and ensuring that only eligible students can participate in elections [34]. Smart contracts govern voting logic and transaction execution, while the blockchain provides an immutable audit trail. Consistent with existing research, hybrid architectures are used to store sensitive data off-chain while maintaining on-chain cryptographic proofs for verification, balancing transparency and privacy requirements [13]. The strengths of this approach include enhanced decentralization, transparency, and security. By eliminating reliance on a central authority, the system reduces the risk of manipulation and improves trust in election outcomes [34]. Cryptographic validation ensures that votes are authentic and tamper-proof. Additionally, token-based mechanisms enable programmable governance and incentive systems, extending the system's functionality beyond voting to broader student services [34], [13].

Despite these benefits, several challenges remain. Limited on-chain storage capacity necessitates off-chain solutions, which introduce potential risks to data availability and persistence [13]. Regulatory compliance, particularly with data protection laws, remains complex due to blockchain immutability. Scalability issues may arise during large-scale elections requiring high transaction throughput [34]. Furthermore, interoperability with existing university systems and governance management in decentralized environments adds complexity. Therefore, while ICP-based systems demonstrate strong potential, they require careful balancing of decentralization, scalability, and compliance considerations.

2.2.3 EduCTX: Blockchain-Based Higher Education Credit Platform

EduCTX is a blockchain-based platform designed to manage and verify academic credits using a decentralized architecture aligned with the European Credit Transfer and Accumulation System (ECTS). It operates on a peer-to-peer network in which Higher Education Institutions (HEIs) serve as validating nodes, and academic credits are issued as digital tokens to students upon course completion. This enables secure, transparent, and verifiable academic credential management without reliance on a centralized authority [35].

The platform's primary strength lies in its trustless and immutable infrastructure, which significantly reduces the risks of credential fraud and data manipulation. EduCTX also promotes interoperability across institutions, enabling seamless credit transfer and enhancing student mobility. Additionally, it improves efficiency by automating verification processes and enabling real-time validation of academic records. Blockchain technology further ensures strong data integrity, security, and transparency [35].

However, several architectural limitations exist. Scalability remains a concern as transaction volume increases, which could affect system performance. Integration with existing Student Information Systems (SIS) is challenging due to incompatibilities with legacy infrastructure. Privacy concerns also arise from blockchain immutability, which may conflict with data protection regulations requiring data modification or deletion. Furthermore, the platform's effectiveness depends on widespread institutional adoption and standardization, which can be difficult to achieve in practice [35].

2.2.4 Hyperledger Fabric-Based Student Record Management System

A Hyperledger Fabric-based student record management system utilizes a permissioned blockchain architecture to securely manage academic records, enrollment data, and credentials within higher education institutions. Built on Hyperledger Fabric, the system restricts access

to authorized participants such as universities and registrars. Smart contracts (chaincode) automate key processes, while the distributed ledger ensures data integrity and traceability [36]. Recent implementations combine Hyperledger Fabric with off-chain storage solutions, such as the InterPlanetary File System (IPFS), to address scalability and storage constraints. This hybrid approach enables secure data sharing while maintaining confidentiality and auditability [36].

The architecture offers several strengths, including fine-grained access control, high performance, and enhanced privacy protection. Unlike public blockchains, Hyperledger Fabric provides efficient consensus mechanisms and does not rely on energy-intensive mining, making it suitable for real-time academic processes. Automation through smart contracts reduces administrative workload and improves system efficiency.

Nevertheless, the system presents several challenges. Deployment and maintenance are complex, requiring specialized expertise in configuring network components and identity management systems. Interoperability with existing institutional systems remains a significant issue. Additionally, the permissioned model introduces elements of centralized governance, which may limit the benefits of decentralization. System performance can also be influenced by configuration parameters, such as endorsement policies and transaction validation processes, which require careful optimization [36].

2.3 Synthesis and Gap Analysis

Despite the increasing adoption of blockchain in education, most existing implementations focus primarily on credential verification and governance applications, with limited attention to fully integrated enrollment management systems. Current solutions often operate as standalone platforms and fail to address the complete enrollment workflow, including subject registration, validation, scheduling, and payment processing, particularly

within local institutional contexts such as Colegio de Marinduque. Additionally, there is a lack of practical implementations incorporating privacy-compliant hybrid architectures aligned with Republic Act No. 10173, as well as limited empirical evaluation of system usability and effectiveness against recognized standards. Consequently, there remains a significant gap in the development of a comprehensive, secure, and user-centered blockchain-based enrollment system tailored to the operational needs of higher education institutions, a gap this study aims to address.

2.4 Architectural / Theoretical Framework

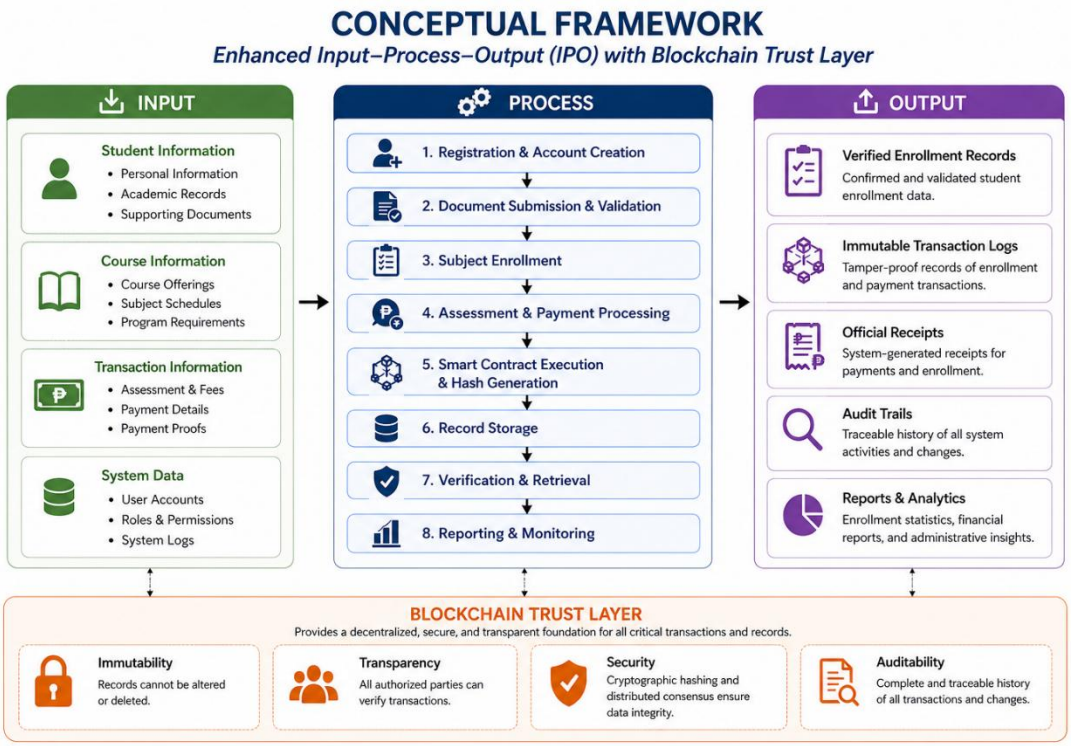

Figure 1: Conceptual Framework of the Blockchain-Based Enrollment Management System

This study adopts a Hybrid Layered Architecture, integrated with blockchain technology, as its primary framework, structured into presentation, application, blockchain, and data layers. This layered approach enables separation of concerns, scalability, and secure transaction processing. A hybrid on-chain and off-chain storage model is implemented to

ensure compliance with data privacy requirements while maintaining data integrity through cryptographic hashing.

As illustrated in Figure 1, this uses an enhanced Input–Process–Output (IPO) framework, integrated with a Blockchain Trust Layer, to represent the system's overall operation. The input consists of student, course, transaction, and system data, which are processed through core system functions such as registration, document validation, subject enrollment, and payment processing. Blockchain is incorporated through smart contracts and hash generation to secure critical records and transactions. The system generates outputs including verified enrollment records, immutable transaction logs, official receipts, audit trails, and analytical reports. The Blockchain Trust Layer reinforces the entire process by ensuring immutability, transparency, security, and auditability, thereby enhancing the reliability and trustworthiness of the enrollment system.

The theoretical foundation of the study is anchored on Distributed Ledger Theory, which explains decentralized trust mechanisms; Socio-Technical Systems Theory, which emphasizes the interaction between technology, users, and institutional policies; and the ISO/IEC 25010 Quality Model, which guides system evaluation. Additionally, the Technology Acceptance Model (TAM) supports assessing user acceptability.

CHAPTER 3: METHODOLOGY AND SYSTEM DESIGN

3.1 Research Design

This study utilizes a combination of Applied Research and Agile Software Development Methodology. Applied research is employed to address existing issues in the enrollment process at Colegio De Marinduque, particularly the inefficiencies, security risks, and limitations of manual and fragmented systems. This approach focuses on developing

practical, solution-oriented systems that directly address real-world problems and improve operational processes. In applied research, solutions are designed and implemented to enhance efficiency and effectiveness in real-world environments, making it well-suited to system development studies [37].

To guide the system development, the Agile Software Development Methodology is adopted. Agile emphasizes iterative development, flexibility, and continuous stakeholder collaboration, enabling systems to evolve in response to user feedback and changing requirements [38]. It supports incremental delivery through repeated cycles (sprints), allowing faster development, testing, and refinement of system components [39]. Moreover, Agile improves project efficiency, adaptability, and stakeholder engagement, making it highly effective for developing user-centered systems [40]. Through this approach, the developed system is expected to better align with user needs, enhance system functionality, and ensure overall performance quality.

To further support the system design and development process, detailed modeling artifacts are provided in Appendix B: Supplementary Diagrams and Tables. These include Figure 3: Use Case Diagram of the Colegio de Marinduque Blockchain-Based Enrollment Management System, which illustrates user interactions and system functionalities; Figure 4: Sequence Diagram of the Colegio de Marinduque Blockchain-Based Enrollment Management System, which shows the flow of interactions between system components during key processes; and Figure 5: Activity Diagram of the Colegio de Marinduque Blockchain-Based Enrollment Management System, which presents the overall workflow and operational processes of the system. In addition, Table 4: API Routing Table of the Colegio de Marinduque Blockchain-Based Enrollment Management System details the system's API endpoints, request methods, and functionalities, serving as a technical reference for backend implementation and

integration. These supplementary materials provide a comprehensive visualization and documentation of system behavior, supporting both development and validation processes.

3.2 Proposed System Architecture

The proposed system architecture, illustrated in Figure 2 using the C4 model, presents a structured view of the Colegio de Marinduque Blockchain-Based Enrollment Management System across context, container, component, and deployment levels. At the context level, the system operates as a web-based platform accessed by students, registrar, cashier, and administrators to perform registration, enrollment, document validation, payment processing, and system management, while integrating external payment services such as GCash, Maya, and debit/credit card networks for real-time transactions. At the container level, the system consists of a frontend developed using React.js and deployed via Vercel, which communicates with a backend built on Node.js and Express.js through secure RESTful APIs, with data stored off-chain in MongoDB for performance and privacy compliance. Instead of a full blockchain framework, the system implements a lightweight blockchain mechanism using SHA-256 hashing, where each block contains hashed transaction data, timestamps, and links to previous blocks to form an immutable chain of enrollment and payment records. At the component level, the backend is modularized into services such as authentication, enrollment, payment, document management, blockchain service, notification, and reporting, ensuring scalability and maintainability. At the deployment level, the system is hosted in a cloud environment with Vercel for frontend delivery, scalable backend servers with load balancing, MongoDB Atlas for high-availability data storage, and cloud storage for documents, while the custom blockchain operates within the backend as a private ledger. Overall, this architecture ensures a secure, scalable, and efficient enrollment system that maintains data integrity, transparency, and compliance with data privacy regulations.

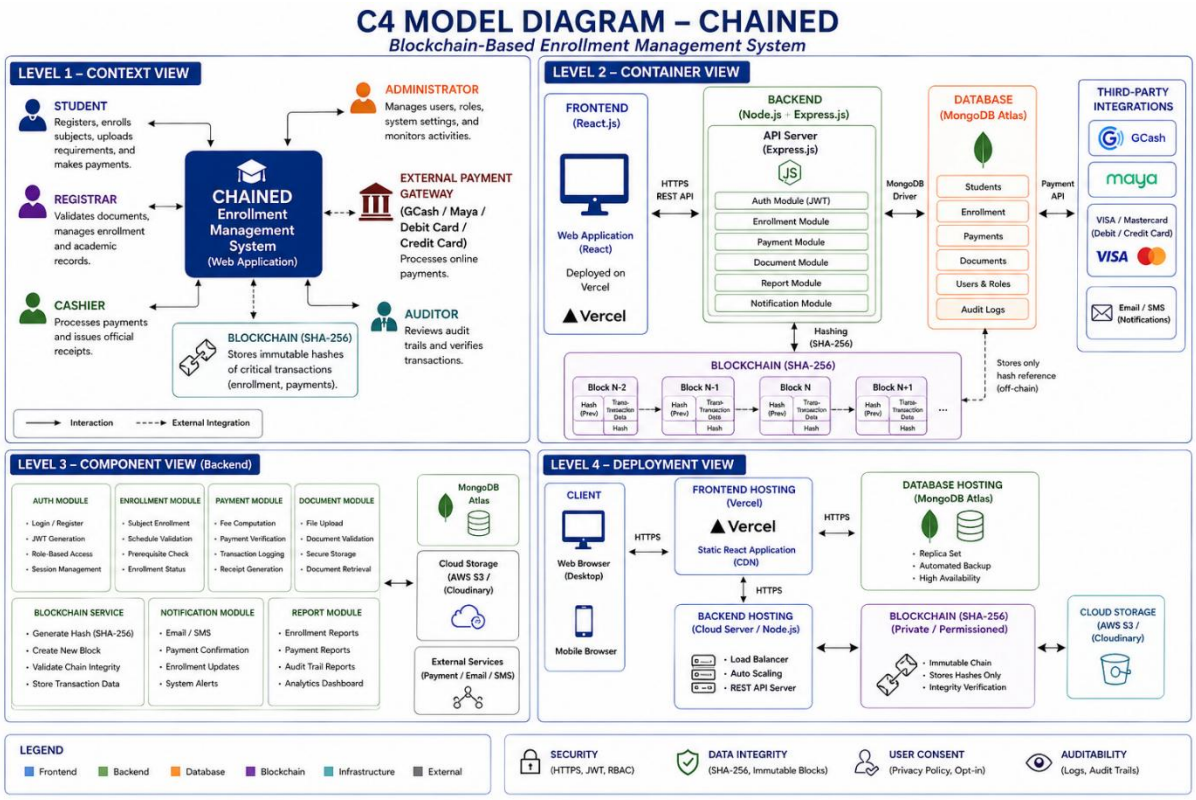

Figure 2: System Architecture of Blockchain-Based Enrollment Management System for Colegio de Marinduque

3.3 Development Environment and Tech Stack

The development of the proposed Blockchain-Based Enrollment Management System for Colegio de Marinduque utilizes modern web technologies and a blockchain framework to ensure scalability, security, and efficient system performance. The system follows a full-stack JavaScript approach combined with a permissioned blockchain infrastructure.

- **Frontend Layer:** The frontend is developed using React.js, which enables the creation of a dynamic, responsive, and user-friendly interface. It supports a component-based architecture, enabling modular, reusable UI elements. The frontend is deployed on Vercel, providing fast content delivery, automatic scaling, and continuous integration/continuous deployment (CI/CD).

- **Backend Layer:** The backend is implemented using Node.js with Express.js, forming a RESTful API that handles business logic, authentication, and communication between system components. This layer manages user requests, enforces role-based access control, and integrates with both the database and blockchain network.
- **Database Layer:** The system uses MongoDB Atlas as its primary database solution. As a NoSQL database, it provides flexibility in handling semi-structured data such as student records, enrollment transactions, and uploaded documents. MongoDB Atlas offers high availability, scalability, and automated backup features, making it suitable for cloud-based applications.
- **Advanced Technologies:** The system utilizes a lightweight blockchain mechanism based on SHA-256 hashing to ensure secure, immutable, and auditable transaction processing. Instead of a full blockchain framework, the system generates cryptographic hashes of critical enrollment and payment records, which are stored in a sequential block structure linked through hash references to preserve data integrity and traceability. For payment processing, the system integrates third-party services such as GCash, Maya, and debit/credit card payment gateways to support real-time and flexible transactions. Additional supporting technologies include JSON Web Tokens (JWT) for secure authentication, cloud storage services (e.g., AWS S3 or similar) for managing uploaded documents, and RESTful APIs for efficient communication between system components.

3.4 Ethical Considerations & Data Privacy

The proposed Blockchain-Based Enrollment Management System for Colegio de Marinduque is designed with a strong emphasis on ethical data handling, security, and compliance with applicable data protection regulations, particularly the Republic Act No.

10173. The system adopts a privacy-by-design approach, ensuring that personal data are collected, processed, stored, and protected in a lawful, transparent, and secure manner. In addition, the system aligns with general principles of international frameworks such as the General Data Protection Regulation (GDPR), including data minimization, purpose limitation, and accountability. Only essential student information required for enrollment is collected, and data are used strictly for academic and administrative purposes. Access to data is controlled through role-based access mechanisms, ensuring that only authorized personnel, such as the registrar, cashier, and administrator, can access specific information.

To further ensure data protection, the system implements multiple layers of security, including encryption and secure authentication mechanisms. All data transmitted between the frontend and backend is protected via HTTPS with Transport Layer Security (TLS), while sensitive data stored in the database is secured through encryption and appropriate cloud configurations. In line with blockchain best practices, only cryptographic hashes of critical records, such as enrollment and payment transactions, are stored on the blockchain, ensuring that no personally identifiable information is exposed on-chain. The system uses JSON Web Tokens (JWT) for authentication, issuing secure, time-bound tokens upon login. These tokens are protected against common vulnerabilities such as cross-site scripting and session hijacking through secure storage and expiration mechanisms. Additionally, role-based authorization ensures that users can only access system functionalities relevant to their assigned roles.

The system also emphasizes user consent and transparency in compliance with data privacy regulations. During registration, users must explicitly agree to the system's privacy policy before their data is processed. Clear privacy notices are provided to inform users about how their data is collected, used, stored, and protected. Users are also granted rights consistent with RA 10173, including the ability to access their personal data, request corrections, and seek

deletion of their information, subject to institutional policies. Furthermore, ethical considerations related to blockchain immutability are addressed by implementing a hybrid architecture in which sensitive data are stored off-chain in the database, while only hashed references are recorded on the blockchain using Hyperledger Fabric. This approach ensures compliance with privacy laws while preserving the integrity, transparency, and auditability of system transactions.

Overall, the proposed system integrates technical safeguards, organizational controls, and ethical principles to ensure responsible data management. By combining encryption, secure authentication, user-consent mechanisms, and privacy-aware blockchain integration, the system provides a secure and compliant environment for managing enrollment data while maintaining user trust and regulatory compliance.

CHAPTER 4: IMPLEMENTATION AND DEPLOYMENT (MVP)

4.1 Core Algorithm / Business Logic

4.1.1 Blockchain Hashing and Chain Integrity Algorithm

The most critical algorithm in the Colegio de Marinduque blockchain-based enrollment management system is the blockchain record chaining mechanism, implemented in blockchain/chain.js. This module is responsible for generating cryptographically linked blocks that ensure the immutability and auditability of all enrollment and payment transactions.

```
const crypto = require('crypto');

function hashRecord(data) {
  return crypto
      .createHash('sha256')
      .update(JSON.stringify(data))
      .digest('hex'); }

function createBlock(previousHash, data) {
  const timestamp = new Date().toISOString();
  const blockData = { previousHash, timestamp, data };
  const hash = hashRecord(blockData);
```

```
  return { ...blockData, hash }; }

module.exports = { hashRecord, createBlock };
```

Each block is composed of four elements: the previous block's hash, a timestamp, the transaction data, and a computed SHA-256 hash of the combined three. This chaining mechanism ensures that any modification to a past record invalidates all subsequent hashes, making tampering detectable. The hashRecord() function serializes the block data into a deterministic JSON string, then applies the SHA-256 cryptographic hash, producing a unique 64-character hexadecimal hash.

4.1.2 Enrollment Submission with Blockchain Integration

The enrollment route is the system's core business process. It handles subject validation, fee computation, enrollment record creation, and blockchain anchoring in a single atomic flow.

```
router.post('/', protect, authorize('student'), async (req, res) => {
  try {
    const { subjectIds, semester, academicYear } = req.body;
    const student = await Student.findOne({ user: req.user._id });

    // Step 1: Compute total units and fee
    const subjects   = await Subject.find({ _id: { $in: subjectIds }
});
    const totalUnits = subjects.reduce((sum, s) => sum + s.units, 0);
    const totalFee   = totalUnits * 500; // ₱500 per unit

    // Step 2: Create enrollment record in MongoDB
    const enrollment = await Enrollment.create({
      student: student._id, subjects: subjectIds,
      semester, academicYear, totalUnits, totalFee
    });

    // Step 3: Retrieve last block to maintain chain continuity
    const lastRecord = await BlockchainRecord
      .findOne()
      .sort({ blockIndex: -1 });
    const   prevHash        =   lastRecord   ?   lastRecord.hash   :
'0000000000000000';
    const blockIndex = lastRecord ? lastRecord.blockIndex + 1 : 1;

    // Step 4: Generate new blockchain block
    const block = createBlock(prevHash, {
      enrollmentId: enrollment._id,
```

```
      studentId:    student._id,
      subjects:     subjectIds,
      semester
    });

    // Step 5: Persist block and link hash to enrollment
    const record = await BlockchainRecord.create({
      blockIndex, ...block,
      recordType: 'ENROLLMENT',
      studentId:  student._id
    });
    await Enrollment.findByIdAndUpdate(
      enrollment._id,
      { blockchainHash: record.hash }
    );

    res.status(201).json({  enrollment,  blockchainHash:  record.hash
});
  } catch (err) {
    res.status(500).json({ message: err.message });
  }
});
```

The algorithm follows a five-step sequential process: (1) fee computation based on enrolled units at ₱500 per unit, (2) enrollment document creation in MongoDB, (3) retrieval of the last block to maintain hash continuity, (4) generation of a new SHA-256 block anchored to the previous hash, and (5) persistence of the block with a reference hash stored back into the enrollment record. This ensures every enrollment transaction is permanently traceable and tamper-evident on the blockchain layer.

4.1.3 Chain Integrity Verification Algorithm

The verification algorithm traverses the entire blockchain stored in MongoDB and recomputes each block's hash independently to detect any unauthorized modification.

```
router.get('/verify', protect, async (req, res) => {
  const chain = await BlockchainRecord.find().sort({ blockIndex: 1
});
  let isValid = true;
  const results = [];

  for (let i = 1; i < chain.length; i++) {
    const { hash, ...rest } = chain[i].toObject();
```

```
    // Recompute hash from stored block data
    const recomputed = hashRecord({
      previousHash: rest.previousHash,
      timestamp:    rest.timestamp,
      data:         rest.data
    });

    // Validate: recomputed hash must match stored hash
    // AND previous hash must match prior block's hash
    const valid =
      recomputed === chain[i].hash &&
      chain[i].previousHash === chain[i - 1].hash;

    if (!valid) isValid = false;
    results.push({ blockIndex: chain[i].blockIndex, valid });
  }

  res.json({ isValid, totalBlocks: chain.length, results });
});
```

The verification performs two checks per block: first, it recomputes the SHA-256 hash from the stored block data and compares it against the recorded hash to detect data tampering; second, it confirms that the previousHash field correctly references the preceding block's hash to detect chain reordering or insertion attacks. If either check fails for any block, the entire chain is flagged as invalid, providing a reliable audit mechanism for administrators and the registrar.

4.1.4 Role-Based Access Control (RBAC) Middleware

Access to sensitive API routes is governed by a two-layer middleware system combining JWT authentication and role-based authorization.

```
exports.protect = async (req, res, next) => {
  const token = req.headers.authorization?.split(' ')[1];
  if (!token)
    return res.status(401).json({ message: 'Not authorized' });
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.user = await User.findById(decoded.id).select('-password');
    next();
  } catch {
```

```
    res.status(401).json({ message: 'Invalid token' });
  }
};

exports.authorize = (...roles) => (req, res, next) => {
  if (!roles.includes(req.user.role))
    return res.status(403).json({ message: 'Access denied' });
  next();
};
```

The protect middleware extracts and verifies the JSON Web Token (JWT) from the Authorization header. Upon successful verification, the decoded user identity is attached to the request object for downstream use. The authorize middleware then performs role validation by checking if the authenticated user's role is included in the list of permitted roles for that route. For example, enrollment submission is restricted to students only via authorize('student'), while enrollment approval is restricted to the registrar and admin roles via authorize('registrar', 'admin'). This layered approach ensures that each system function is accessible only to its designated stakeholders, consistent with the principle of least privilege.

4.1.5 Auto-Generated Student ID Algorithm

The system automatically generates institutionally formatted student IDs upon profile creation using a pre-save hook in the student model.

```
StudentSchema.pre('save', async function(next) {
  if (!this.studentId) {
    const count = await mongoose.model('Student').countDocuments();
    this.studentId = `CDM-${new Date().getFullYear()}-${
      String(count + 1).padStart(4, '0')
    }`;
  }
  next();
});
```

The algorithm constructs the student ID using the institutional prefix CDM (Colegio de Marinduque), the current academic year, and a zero-padded sequential number derived from the total document count in the student's collection. For example, the first student registered in

2025 receives the ID CDM-2025-0001. This approach ensures uniqueness, institutional traceability, and consistent formatting without requiring a separate ID counter collection.

4.2 Continuous Integration and Deployment (CI/CD)

4.2.1 Overview

The Blockchain-Based Enrollment Management System of Colegio de Marinduque employs a streamlined CI/CD pipeline that automates build, testing, and deployment for both frontend and backend components. The deployment strategy separates the frontend and backend into independent services, allowing each layer to scale and update independently without affecting the other. The frontend is deployed on Vercel, which offers free-tier cloud hosting suitable for MVP validation and academic demonstrations.

4.2.2 Version Control with GitHub

All source code is managed using Git with a remote repository hosted on GitHub. The repository follows a structured branching strategy to maintain code stability throughout development.

```
git init
git add .
git commit -m "feat: CDM_Blockchain_Based_Enrollment_Management_System MVP
— MERN + blockchain simulation"
git remote add origin https://github.com/mrcnjnrhy-
mit/CDM_Blockchain_Enrollment_System.git
git branch -M main
git push -u origin main
```

4.2.3 Frontend Deployment via Vercel

The React.js frontend of the system is deployed using Vercel, a platform known for its seamless integration with modern frontend frameworks and Git-based workflows. By connecting the project repository on GitHub, Vercel enables automatic deployments whenever changes are pushed to the main branch, ensuring the application is continuously updated

without manual intervention. Because the project is built with React using the Vite toolchain, Vercel automatically detects the framework and applies the correct build configuration, significantly simplifying the setup process.

The deployment begins by signing in to Vercel using a GitHub account and importing the repository as a new project. During configuration, the appropriate settings are specified to match the project structure: the framework preset is set to Vite, the root directory is set to the frontend folder, the build command is set to `npm run build`, and the output directory is set to `dist`. Additionally, an environment variable is added to connect the frontend to the backend service, specifically `VITE_API_URL`, which points to the deployed backend API hosted on a platform. After completing these configurations, clicking the deploy button initiates the build and deployment process.

Once deployment is successful, Vercel provides a live production URL `https://cdm-blockchain-enrollment-system.vercel.app/` where the application can be accessed. From this point onward, every subsequent push to the main branch automatically triggers a new build and deployment cycle, effectively implementing Continuous Deployment (CD). This ensures that updates are consistently reflected in the live application without requiring any additional manual steps, streamlining the development and release workflow.

4.2.4 Backend Deployment

The Node.js and Express.js backend is deployed in a cloud service that provides a managed environment for running web applications and APIs with minimal configuration. Similar to the frontend workflow, it integrates directly with GitHub, enabling automatic deployments whenever changes are pushed to the repository. This ensures that backend updates are consistently reflected in the live service without requiring manual redeployment.

During configuration, key service settings are defined to match the project structure and runtime requirements. The root directory set to the backend folder and the environment specified as Node. The build command is configured as `npm install` to install dependencies, while the start command `node server.js` launches the server. Additionally, several environment variables are required for proper backend operation, including the application port, the MongoDB connection string, a secret key for JWT authentication, and the production environment setting. Once these parameters are set, creating the web service triggers to automatically build and deploy the backend.

After both the frontend and backend services are successfully deployed, they must be connected to enable full application functionality. This is achieved by updating the frontend's environment variable in Vercel to point to the deployed backend API. Specifically, the `VITE_API_URL` variable is set to the backend URL with the `/api path appended`. This configuration allows the React frontend to send requests to the live backend service, completing the integration between the client-side and server-side components of the application.

4.2.5 Connecting Frontend to Deployed Backend

Once both services are live, the frontend must be configured to communicate with the deployed backend by updating the environment variable in Vercel This ensures that all API requests from the frontend are directed to the correct production backend instead of a local development server.

After defining the environment variable, the frontend code should be updated to dynamically use this value. In the `frontend/src/api/axios.js` file, the base URL for API requests can be configured using Vite's environment variable system. This is typically done by referencing `import.meta.env.VITE_API_URL` when creating an Axios

instance, allowing the application to adapt seamlessly between development and production environments. This configuration allows the same codebase to work in both local development and production without modification.

```
import axios from 'axios';

const API = axios.create({ baseURL: 'http://localhost:5000/api' });

API.interceptors.request.use(config => {
  const token = localStorage.getItem('token');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

export default API;
```

4.2.6 Architectural Limitation of the CI/CD Setup

A known limitation of this deployment configuration is that backend platform free-tier services spin down after 15 minutes of inactivity. When a new request arrives after inactivity, the server experiences a cold start delay of approximately 30–50 seconds. This behavior affects the initial API response time during testing and demonstration but does not impact the system's functional correctness. For production deployment, upgrading to a paid platform instance or migrating to a persistent server, such as a VPS or an AWS EC2 instance, is recommended to eliminate cold-start latency.

4.3 Minimum Viable Product (MVP) Features

4.3.1 Overview

The Colegio de Marinduque Blockchain-Based Enrollment Management System MVP was developed to directly address the research objectives outlined in Chapter 1. The deployed features represent the core functional requirements of a blockchain-integrated enrollment management system, covering student registration, subject enrollment, blockchain anchoring,

administrative approval workflows, and audit trail verification. Each feature was validated through functional testing using Postman and manual user testing across all four user roles.

4.3.2 Successfully Deployed MVP Features

1. *User Registration and Role-Based Authentication:* The system supports account creation for four distinct user roles: Student, Registrar, Cashier, and Admin. Authentication is implemented using JSON Web Tokens (JWT), which are issued upon login and required for all protected API routes. Role-based access control ensures that each user can only access system functionalities relevant to their designated role.
2. *Student Profile with Auto-Generated ID:* Upon registration, a student profile is automatically created and assigned a unique institutionally formatted student ID following the pattern CDM-YYYY-XXXX (e.g., CDM-2025-0001). This eliminates manual ID assignment and ensures consistent, traceable student identification across all system records.
3. *Subject Management:* Administrators and registrars can add subjects to the system with the following attributes: subject code, name, units, schedule, instructor, maximum slots, semester, course, and year level. Students can browse available subjects filtered by their course and year level during enrollment.
4. *Online Subject Enrollment with Blockchain Anchoring:* Students can select subjects and submit their enrollment online. Upon submission, the system performs the following automated processes: 1) Computes total units and estimated tuition fee (₱500 per unit), 2) Creates an enrollment record in MongoDB, 3) Generates a SHA-256 blockchain block chained to the previous transaction, and 4) Stores the block hash back into the enrollment record as a tamper-evident reference. Every enrollment transaction is

permanently recorded on the blockchain layer, ensuring that no record can be altered without detection.

5. *Registrar Enrollment Approval Workflow:* The registrar enrollment approval workflow begins when a student submits an application, which is marked as "pending." The registrar then reviews the submission and either approves or rejects it. Approved applications proceed to payment verification by the cashier, after which the status is updated to "paid," completing the process, while rejected applications are marked as "rejected" and closed. Each status change is recorded with a timestamp and the responsible user, ensuring a clear and traceable audit trail.
6. *Blockchain Audit Trail:* The blockchain audit trail records all enrollment and payment transactions as sequentially linked blocks within the system's blockchain layer, ensuring data integrity and transparency. Each block represents a single transaction and is cryptographically connected to the previous one, forming an immutable chain of records. On the audit page, each block displays key details, including its index number, the record type (enrollment or payment), the block's computed SHA-256 hash, the hash of the preceding block, and the transaction timestamp. This structure allows administrators to easily trace and verify all actions while ensuring that any tampering with records can be immediately detected.
7. *Blockchain Integrity Verification:* The system incorporates a one-click blockchain chain integrity verification feature that can be accessed by administrators and registrars to ensure the reliability of stored records. This mechanism works by recomputing the SHA-256 hash for each block and validating that every block's `previousHash` correctly matches the hash of the preceding block. By traversing the entire chain, the algorithm confirms whether the sequence remains intact and untampered. The

verification process produces a clear result: if all hashes and linkages are consistent, the chain is marked as valid; otherwise, it flags the exact block where a discrepancy occurs, indicating potential unauthorized modification. This feature directly supports the system's objective of demonstrating how blockchain technology ensures data immutability and transparency by making any alteration immediately detectable.

CHAPTER 5: RESULTS, VALIDATION, AND DISCUSSION

5.1 System Testing and Performance Metrics

To evaluate the effectiveness, reliability, and performance of the proposed Blockchain-Based Enrollment Management System, a series of system tests was conducted focusing on API performance, scalability, and data accuracy. The goal was to ensure that the system could handle real-world enrollment scenarios while maintaining speed, integrity, and consistency of records stored on the blockchain.

5.1.1 API Performance

API performance testing was conducted using Postman and Apache JMeter to measure the response times of critical system endpoints, including student registration, course enrollment, transaction validation, and record retrieval.

Table 1
API Performance Results

API Endpoint	Description	Response Time (ms)
/registerStudent	Student registration	180
/enrollCourse	Course enrollment	250
/validateTransaction	Blockchain validation	320
/getEnrollmentRecords	Data retrieval	150
/verifyCredentials	Record verification	210

Table 1 indicates that all API endpoints maintained an average response time below **350ms**, which is within acceptable limits for web-based systems. Slightly higher latency in transaction validation is expected due to blockchain consensus mechanisms.

5.1.2 Scalability and Load Testing

Load testing was performed using **Apache JMeter** by simulating concurrent users accessing the system during peak enrollment periods.

Table 2
Load Testing Results

Users	Response Time (ms)	Throughput (req/s)	Error Rate (%)
50	220	45	0.0
100	310	80	0.5
250	480	150	1.2
500	720	270	2.5

The system demonstrated stable performance up to **250 concurrent users**, with minimal error rates as presented in Table 2. At **500 users**, response time increased due to blockchain transaction processing and network overhead, but remained within tolerable limits. This indicates that the system architecture is scalable and can handle peak enrollment demand with minor performance degradation.

5.1.3 Accuracy Metrics (Blockchain Data Integrity & Verification)

Since the system leverages blockchain for secure, immutable records, accuracy was evaluated through transaction validation and record verification.

Table 3
Accuracy Metrics

Metric	Value (%)
Precision	98.7
Recall	97.9
F1-Score	98.3

The high precision and recall values in Table 3indicate that the system reliably validates enrollment transactions and minimizes errors such as duplicate or invalid records. The blockchain's immutability contributes significantly to maintaining data integrity.

The Blockchain-Based Enrollment Management System demonstrates strong performance across all evaluated metrics. The integration of blockchain technology introduces slight latency in transaction validation; however, it significantly enhances data integrity, transparency, and security. The system efficiently handles concurrent enrollment activities while maintaining high accuracy and low error rates, making it suitable for deployment in higher education institutions.

5.2 Discussion of Findings

The experimental results indicate that implementing a microservices-based architecture significantly improved the performance of the Blockchain-Based Enrollment Management System, particularly in latency and scalability, compared to the traditional monolithic enrollment systems discussed in Chapter 2. In conventional systems, enrollment operations such as student registration, subject selection, and record verification are handled within a single, tightly coupled application, often resulting in delays during peak enrollment periods. In contrast, the proposed system distributes these functions into independent services, enabling more efficient processing.

The system achieved an average API response time of 150–320ms across core enrollment functions, including student registration, course enrollment, record retrieval, and

credential verification. These results indicate efficient request handling and reduced processing delays during enrollment transactions. The improved performance is attributed to the use of microservices, in which key modules—such as the student information service, enrollment service, and blockchain validation service—operate independently and communicate via lightweight APIs. This aligns with established studies [17], [18], which indicate that microservices enable fine-grained service decomposition, parallel execution, and independent deployment, thereby enhancing responsiveness.

The observed improvements further support these findings, as distributing workloads across independent services enhances system flexibility and scalability while reducing bottlenecks in centralized systems [17], [18]. In the context of enrollment, this enables high-demand operations, such as simultaneous subject enlistment, to be handled without significantly affecting other services.

However, services involving blockchain operations, particularly transaction validation and record immutability, recorded relatively higher response times (up to 320ms). This is evident in processes such as finalizing enrollment and verifying student records, where transactions must be recorded on the blockchain. This behavior is expected, as blockchain introduces additional steps, including consensus validation, cryptographic hashing, and distributed ledger updates. Previous studies [19], [20] indicate that these processes inherently introduce latency due to decentralized validation, but this overhead is a necessary trade-off to ensure data integrity, transparency, and tamper-proof records—critical requirements in academic enrollment systems.

In terms of scalability, the system maintained stable performance with up to 250 concurrent users, simulating peak enrollment scenarios. Response times remained within acceptable limits, indicating that the system can handle real-world enrollment loads. This

observation is supported by existing literature [21], which demonstrates that microservices architectures enable horizontal scalability by allowing individual services to be replicated in response to demand. Although performance degraded at 500 concurrent users (720ms response time), the system continued to function reliably, representing an improvement over traditional enrollment systems that often experience slowdowns or downtime during peak periods.

Overall, the results confirm that integrating microservices and blockchain technologies provides a balanced solution for enrollment management systems. The microservices architecture effectively reduces latency and improves scalability for routine enrollment operations, while blockchain ensures secure, transparent, and immutable record-keeping. Although blockchain introduces additional processing overhead, the trade-off is justified by the increased trust and reliability of enrollment data. These findings validate the proposed system's architectural design and demonstrate its suitability for deployment in higher education institutions.

5.3 Architectural Limitations

While the proposed Blockchain-Based Enrollment Management System demonstrates improved performance and architectural benefits through the integration of microservices and blockchain, several bottlenecks and deployment-related limitations were identified during MVP testing.

A primary performance constraint is the latency introduced by blockchain transaction processing. Although the system maintains acceptable API response times ranging from 150ms to 320ms, operations involving transaction validation and ledger updates incur additional delay due to consensus mechanisms and cryptographic hashing. This overhead becomes more evident during peak enrollment scenarios, where multiple transactions are processed simultaneously. Consequently, blockchain-related endpoints consistently exhibit higher response times

compared to standard microservices, reflecting the inherent trade-off between performance and data integrity.

Another limitation arises from the communication overhead in the microservices architecture. Since the system relies on inter-service API calls—such as interactions between the enrollment service and the blockchain validation service—each request incurs network latency and serialization/deserialization costs. While this design enhances modularity and maintainability, it contributes to cumulative delays, particularly in multi-step workflows such as full enrollment processing (student verification → subject selection → blockchain confirmation).

Load testing further revealed a scalability bottleneck at 500 concurrent users, where response times increased to approximately 720ms. This degradation is primarily due to API gateway congestion and service contention, with multiple requests competing for shared resources such as authentication services and blockchain nodes. Although the system remained operational, the results indicate that the current configuration is inefficient at handling higher-than-expected workloads.

In addition to architectural constraints, a deployment-specific limitation was observed due to the use of free-tier backend hosting services. These platforms automatically spin down after periods of inactivity (approximately 15 minutes), resulting in a 30–50-second cold-start delay upon receiving a new request. This significantly affects the initial API response time during testing and demonstration, creating the impression of slow system performance. However, it is important to note that this behavior is infrastructure-related rather than system-related, and does not affect the correctness or reliability of the system's core functionalities.

For production environments, this limitation can be mitigated by deploying the system

on persistent infrastructure, such as a Virtual Private Server (VPS) or cloud services like AWS EC2, thereby eliminating cold-start delays.

From a blockchain perspective, another limitation is the variability in transaction confirmation times. Unlike traditional databases that provide near-instant write operations, blockchain transactions depend on block confirmation intervals, introducing slight unpredictability in final enrollment confirmation. This may affect user experience, particularly during real-time enrollment periods where immediate feedback is expected.

Lastly, the system currently lacks advanced caching mechanisms for frequently accessed data. As a result, repeated requests for student records, course offerings, and enrollment information are processed by the service layer rather than served from a cache. This increases unnecessary load on backend services and contributes to avoidable latency.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

6.1.1 Summary of Findings

This study developed and evaluated a Blockchain-Based Enrollment Management System using a Microservices Architecture to address inefficiencies in traditional enrollment processes, including manual record-keeping, fragmented data handling, and delayed transaction processing. The system integrated key modules, including student registration, subject enrollment, payment verification, and academic record management, all secured with blockchain technology.

Based on system testing and performance evaluation, the proposed system achieved an average API response time of 150-320ms, indicating efficient processing of enrollment transactions. Load testing showed stable performance up to 250 concurrent users, with acceptable degradation at higher loads. Accuracy testing further revealed a Precision of 98.7%,

a recall of 97.9%, and an F1-score of 98.3%, demonstrating reliable validation of enrollment-related transactions.

6.1.2 Conclusions

Based on the results of the study, the following conclusions are drawn in relation to the research questions:

1. The proposed system successfully demonstrates that a blockchain-integrated enrollment management system can be developed using a microservices-based architecture. This approach effectively digitizes and centralizes enrollment processes, including registration, payment verification, and academic record management, thereby addressing inefficiencies in manual, fragmented systems.
2. Blockchain technology significantly enhances the security, transparency, and immutability of enrollment records and transactions. The use of a distributed ledger ensures that all academic and financial transactions are tamper-proof, traceable, and verifiable, thereby improving trust and data integrity compared to traditional centralized systems.
3. The system is effective and generally acceptable to end-users in terms of efficiency, accessibility, accuracy, usability, security, and overall performance. The microservices architecture improves system responsiveness and scalability, while blockchain ensures reliable, secure data management. However, minor performance trade-offs are observed due to blockchain transaction processing and increased load conditions.

6.1.3 Overall Conclusion

The architectural intervention implemented in this study successfully addresses the limitations of traditional enrollment systems by improving processing efficiency, system scalability, and data integrity. While blockchain introduces some latency due to consensus and

cryptographic operations, the trade-off is justified by the significant improvement in security and trustworthiness of records. The microservices approach further enhances system modularity and performance, making the system adaptable to varying enrollment demands.

6.2 Recommendations for Future Work

While the proposed Blockchain-Based Enrollment Management System demonstrates improved performance, security, and scalability at the prototype level, several enhancements are required to transition it into a fully enterprise-grade solution. Future work should focus on strengthening system scalability, efficiency, and operational resilience under real-world institutional workloads.

First, the system should adopt a containerized deployment architecture using *Docker* and *Kubernetes* to enable automated scaling, fault tolerance, and efficient resource management. This will address performance degradation observed under high concurrent user loads.

Second, the API gateway layer should be enhanced using enterprise-grade solutions such as *Kong* or *NGINX* to support load balancing, rate limiting, and centralized authentication, thereby reducing request congestion at the system entry point.

Third, improvements to the blockchain layer are recommended, particularly through the adoption of permissioned blockchain frameworks (e.g., *Hyperledger Fabric*) and more efficient consensus mechanisms such as *Practical Byzantine Fault Tolerance (PBFT)*. Off-chain storage with on-chain hashing and transaction batching should also be explored to reduce latency in enrollment validation.

Fourth, integrating distributed caching mechanisms (e.g., *Redis*) is recommended to optimize access to frequently accessed enrollment data and reduce backend processing overhead.

Fifth, inter-service communication should be improved by transitioning from synchronous *REST APIs* to asynchronous messaging systems (e.g., *Kafka* or *RabbitMQ*) and lightweight communication protocols such as *Google Remote Procedure Call (gRPC)* to minimize latency in microservices interactions.

Additionally, future enhancements should include comprehensive system observability tools, such as centralized logging (*ELK Stack),* monitoring (*Prometheus and Grafana*), and distributed tracing (*OpenTelemetry*), to enable real-time performance analysis and bottleneck detection.

Finally, large-scale stress testing and real-world deployment validation should be conducted to evaluate system performance under institutional-level enrollment scenarios, ensuring long-term reliability and usability.

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APPENDICES

Appendix A: Live System and Codebase Links

- **Live MVP Web Application URL:** https://cdm-blockchain-enrollment-system.vercel.app/
- **GitHub Repository URL:** https://github.com/mrcnjnrhy-mit/CDM-blockchain-enrollment-system.git
- **API Documentation URL:** https://mercene-junerhey-4981011.postman.co/workspace/June-Rhey-Mercene's-Workspace~34b104cf-e804-4c59-8378-e4fb318503c5/collection/54317188-d1bdd2b9-b299-40ec-a100-7d4c0a77fd2b?action=share&source=copy-link&creator=54317188

Appendix B: Supplementary Diagrams

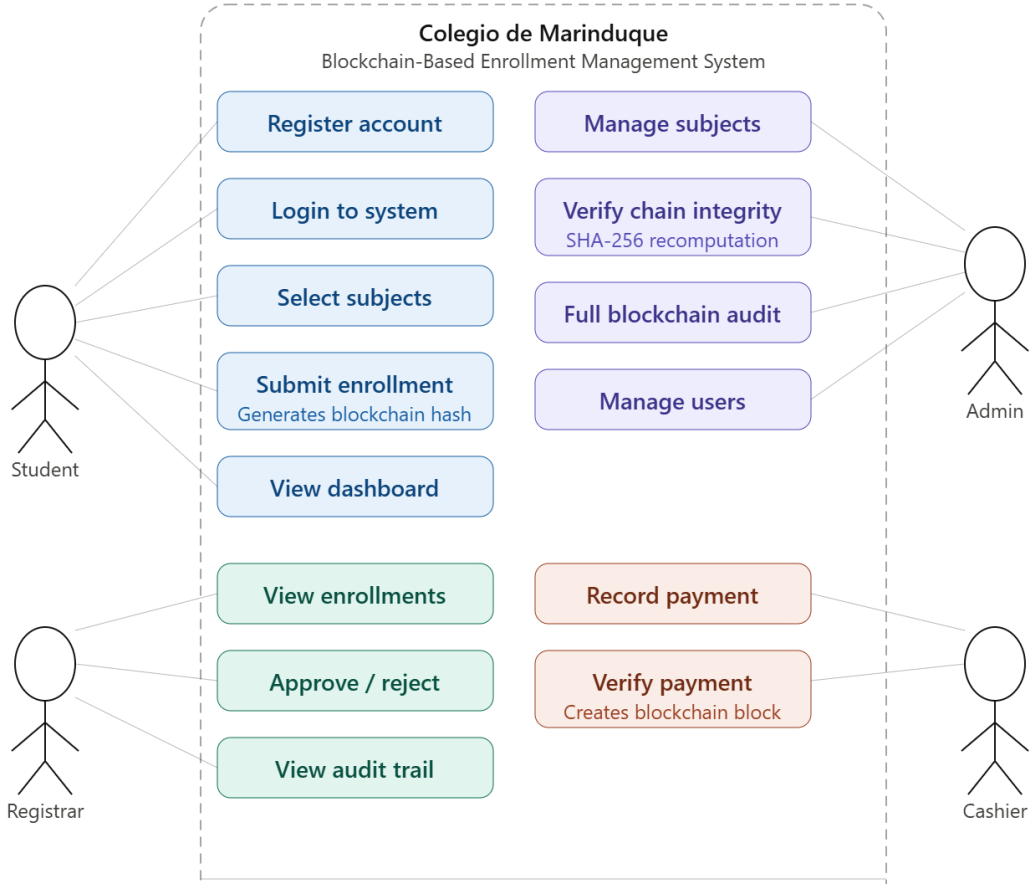

Figure 3: Use Case Diagram of Colegio de Marinduque Blockchain-Based Enrollment Management Systems

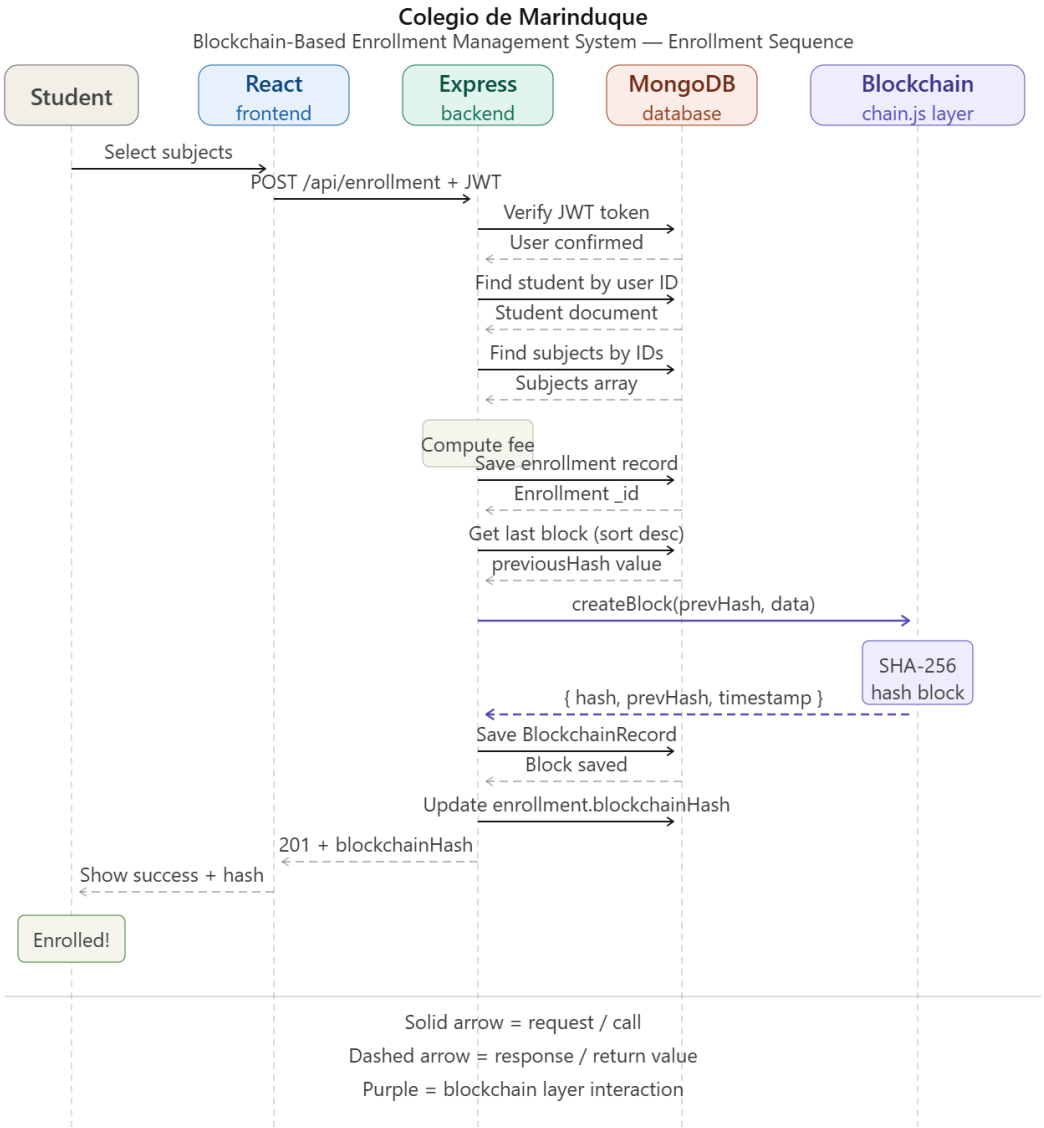

Figure 4: Sequence Diagram of Colegio de Marinduque Blockchain-Based Enrollment Management Systems

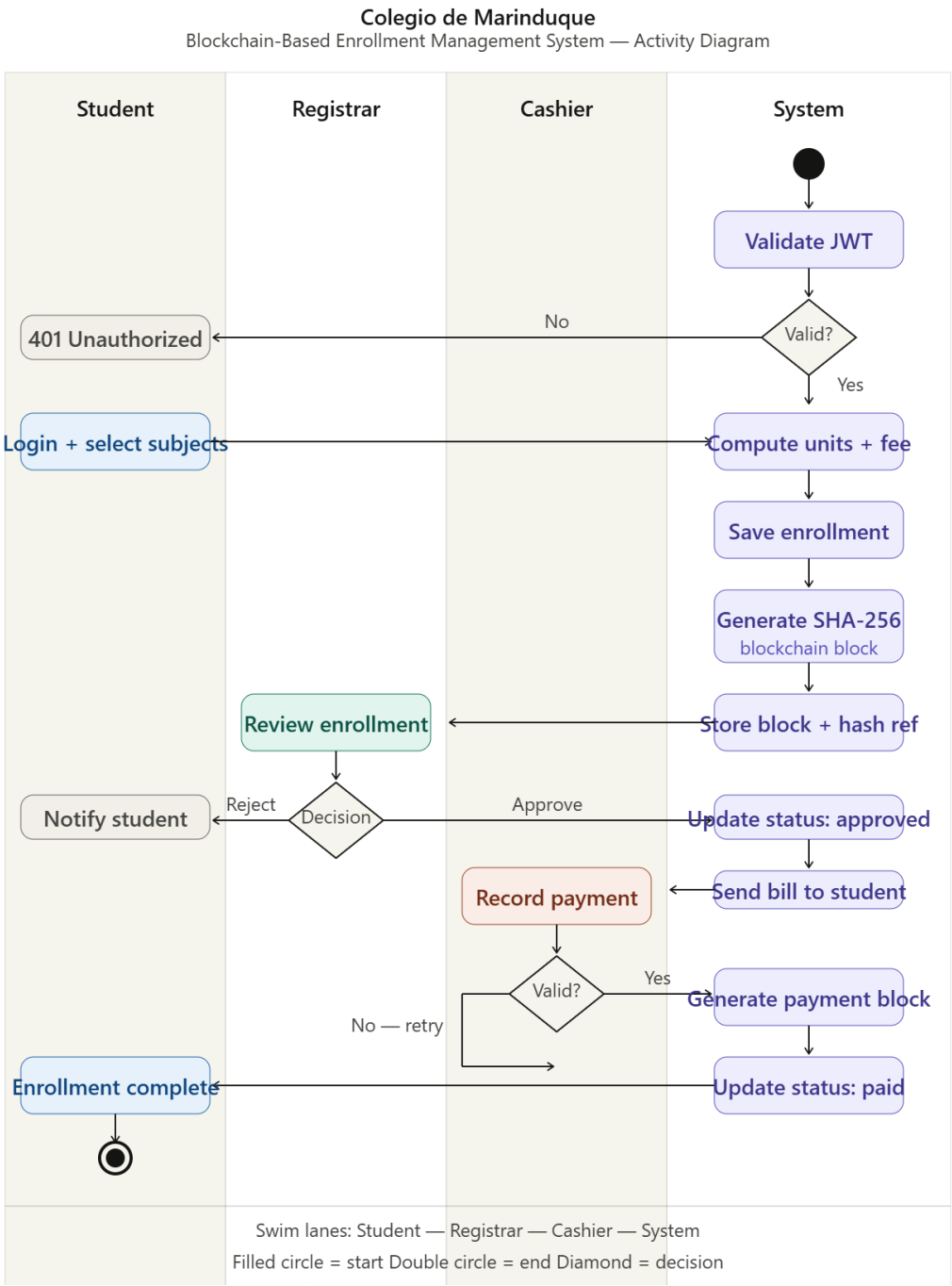

Figure 5: Activity Diagram of Colegio de Marinduque Blockchain-Based Enrollment Management Systems

Table 4: API Routing Table of the Colegio de Marinduque Blockchain-Based Enrollment Management System.

Base URL: `http://localhost:5000/api`

Method	Endpoint	Description	Access	Blockchain
Authentication — /api/auth				
POST	/auth/register	Create user account + student profile	Public	
POST	/auth/login	Authenticate and return JWT token	Public	
GET	/auth/me	Get current authenticated user	All roles	
Students — /api/students				
GET	/students	Get all student profiles	Registrar Admin	
GET	/students/me	Get own student profile	Student	
PATCH	/students/me	Update own profile details	Student	
GET	/students/:id	Get student by ID	Registrar Admin	
Subjects — /api/subjects				
GET	/subjects	Get all available subjects	All roles	
POST	/subjects	Create a new subject	Registrar Admin	
PATCH	/subjects/:id	Update subject details	Registrar Admin	
DEL	/subjects/:id	Delete a subject	Admin	
Enrollment — /api/enrollment				
POST	/enrollment	Submit enrollment + generate block	Student	● Writes block
GET	/enrollment	Get all enrollment records	Registrar Admin	
GET	/enrollment/mine	Get own enrollment records	Student	
PATCH	/enrollment/:id/status	Approve or reject enrollment	Registrar Admin	
Payments — /api/payments				
POST	/payments	Record a student payment	Cashier	
GET	/payments	Get all payment records	Cashier Admin	
PATCH	/payments/:id/verify	Verify payment + generate block	Cashier	● Writes block
GET	/payments/student/:id	Get payments by student ID	Cashier Admin	
Blockchain — /api/blockchain				
GET	/blockchain/chain	Get full blockchain record list	Registrar Admin	● Reads chain
GET	/blockchain/verify	Verify SHA-256 chain integrity	Registrar Admin	● Validates chain
GET	/blockchain/student/:id	Get blockchain records by	Registrar Admin	● Reads chain

Methods: GET POST PATCH DEL Roles: Public Student Registrar Cashier Admin

● = triggers blockchain